Chapter 6. Potential Outcomes and Effect Estimation

Potential outcomes, IPW and doubly robust estimation, CATE meta-learners, the R-learner

Contents

6.1 Potential outcomes **3**
- 6.1.1 Frameworks for causal inference . . . 3
- 6.1.2 Basic concepts . . . 3
  - 6.1.2.1 The fundamental problem of causal inference . . . 3
  - 6.1.2.2 God's eye view: individual versus average effects . . . 4
  - 6.1.2.3 What the researcher actually sees . . . 4
- 6.1.3 Assumptions linking potential and observed outcomes . . . 4
  - 6.1.3.1 SUTVA . . . 4
  - 6.1.3.2 Consistency and well-defined interventions . . . 5
- 6.1.4 Causal estimands . . . 5
- 6.1.5 Randomization and observational studies . . . 5
  - 6.1.5.1 Unconfounded assignment . . . 5
  - 6.1.5.2 The role of randomization . . . 5
  - 6.1.5.3 Observational studies . . . 6
  - 6.1.5.4 Identification of the ATE . . . 6
  - 6.1.5.5 Regression estimation of the ATE . . . 6
  - 6.1.5.6 Summary . . . 6
- 6.1.6 Appendix: Positivity and consistency in causal diagrams . . . 6
- 6.1.7 Appendix: Confounding and exchangeability . . . 7

6.2 Inverse probability weighting and doubly robust estimation **7**
- 6.2.1 The propensity score . . . 7
- 6.2.2 Inverse probability weighting . . . 8
  - 6.2.2.1 The IPW identity . . . 8
  - 6.2.2.2 Estimators . . . 8
  - 6.2.2.3 Normalizing the weights . . . 8
  - 6.2.2.4 The weighted population . . . 8
  - 6.2.2.5 Pros and cons . . . 8
  - 6.2.2.6 Trimming and truncation . . . 9
  - 6.2.2.7 PS weighting in randomized trials . . . 9
  - 6.2.2.8 Weighting versus balancing . . . 9
- 6.2.3 Doubly robust estimation . . . 9
  - 6.2.3.1 Terminology . . . 9
  - 6.2.3.2 Two building blocks . . . 9
  - 6.2.3.3 Verification with the true nuisance functions . . . 10
  - 6.2.3.4 Two equivalent forms . . . 10
  - 6.2.3.5 What does it estimate? . . . 10
  - 6.2.3.6 The doubly robust ATE estimator . . . 10
  - 6.2.3.7 Concluding remarks . . . 11
  - 6.2.3.8 Summary . . . 11

6.3 Conditional average treatment effects **11**
- 6.3.1 Why heterogeneous treatment effects? . . . 11
  - 6.3.1.1 The average can be misleading . . . 11
  - 6.3.1.2 Effect modification . . . 11
  - 6.3.1.3 Why care? . . . 12
  - 6.3.1.4 From stratification to CATE estimation . . . 12

6.3.2 Meta-learners for CATE . . . 12
6.3.2.1 T-learner and S-learner . . . 12
6.3.2.2 Regression imputation . . . 12
6.3.2.3 X-learner . . . 12
6.3.2.4 DR-learner . . . 13

6.4 The R-learner 13
6.4.1 Motivation and identification . . . 13
6.4.2 The R-loss . . . 14
6.4.3 Algorithm . . . 14
6.4.4 Orthogonality and the quasi-oracle property . . . 14
6.4.5 Comparison and summary . . . 15

6.1 Potential outcomes

Motivation. Define and identify causal effects as comparisons of potential outcomes, a missing-data problem, instead of structural equations and graphs.

6.1.1 Frameworks for causal inference

Motivation. Place the potential outcome (PO) framework next to the causal-diagram framework.

Two frameworks are in common use:

- the **potential outcome framework** (counterfactual framework, Neyman–Rubin causal model; Neyman 1923, Rubin 1974);
- the **causal diagram framework** (Pearl 2009).

They are mathematically connected (Richardson and Robins 2013; see the SWIG lecture).

The PO framework dominates medicine, policy and the social sciences. Causal inference is a *missing-data problem*, with explicit modeling of the *assignment mechanism* that reveals the data; it grew from Fisher (1918, 1925) and Neyman (1923) on randomized experiments, extended by Rubin (1974, 1978) to non-randomized studies.

6.1.2 Basic concepts

Motivation. Units, treatments, potential outcomes, causal effects, and the fundamental problem.

- **Unit:** the person, place or thing a treatment operates on, at a particular time (the same person at two times is two units).
- **Treatment:** an intervention whose effect on the units is assessed relative to no intervention (control).
- **Potential outcomes:** the unit's measurement under active treatment, $Y(1)$, and under control, $Y(0)$; also *counterfactuals*.
- **Causal effect:** a per-unit comparison of $Y(1)$ and $Y(0)$, e.g. $Y_i(1) - Y_i(0)$.

6.1.2.1 The fundamental problem of causal inference

At most one potential outcome is observed per unit; the other is counterfactual (Holland 1986).

Unit	$Y(x{=}0)$	$Y(x{=}1)$	X	Y
1	?	1	1	1
2	0	?	0	0
3	?	0	1	0
4	?	1	1	1
5	1	?	0	1

Neyman's urn model. Each unit is a ticket carrying Y_1 and Y_0; assignment reveals only Y_1 for treated units and Y_0 for controls.

6.1.2.2 God's eye view: individual versus average effects

Y: death (1) or survival (0), both potential outcomes visible (adapted from Hernán and Robins, *What If*, Table 1.1):

	$Y(0)$	$Y(1)$	effect		$Y(0)$	$Y(1)$	effect
Rheia	1	0	-1	Hestia	1	0	-1
Zeus	0	0	0	Poseidon	1	0	-1
Kronos	0	1	$+1$	Hera	1	0	-1
Demeter	0	0	0	Artemis	1	1	0
Hades	0	0	0	Apollo	0	1	$+1$
Risk: $\mathbb{E}[Y(0)] = 0.5$, $\mathbb{E}[Y(1)] = 0.3$							

The average causal effect is $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = 0.3 - 0.5 = -0.2$, but individual effects vary: 4 helped, 2 harmed, 4 unaffected.

6.1.2.3 What the researcher actually sees

(Adapted from *What If*, Table 1.2.)

	$Y(0)$	$Y(1)$	X	Y^{obs}		$Y(0)$	$Y(1)$	X	Y^{obs}
Rheia	?	0	1	0	Hestia	?	0	1	0
Zeus	0	?	0	0	Poseidon	1	?	0	1
Kronos	?	1	1	1	Hera	?	0	1	0
Demeter	0	?	0	0	Artemis	1	?	0	1
Hades	0	?	0	0	Apollo	?	1	1	1

By consistency $Y^{\text{obs}} = Y(X)$. The naive comparison gives $2/5 = 0.4$ in both groups, a difference $0 \neq -0.2$: sicker patients ($Y(0) = 1$) chose treatment more often (confounding), hence the need for exchangeability.

6.1.3 Assumptions linking potential and observed outcomes

Motivation. SUTVA and consistency, and their failures.

6.1.3.1 SUTVA

Assumption 1 (Stable Unit Treatment Value Assumption; Rubin 1980). (1) *No interference* and (2) *no different versions* of a treatment (e.g. the same surgery by different surgeons). It links the observed X with the intervention x:

$$X_i = 1 \Rightarrow Y_i = Y_i(1), \qquad X_i = 0 \Rightarrow Y_i = Y_i(0),$$

equivalently

$$Y_i = X_i Y_i(1) + (1 - X_i) Y_i(0).$$

Violation by interference: $Y_i(x)$ depends on others' treatments (vaccination, advertising, infectious disease, social networks, agricultural plots); other assumptions are then needed (Rosenbaum 2007; Hudgens and Halloran 2008), and spillovers are often of interest themselves.

On the history of SUTVA (VanderWeele and Hernán, "Causal inference under multiple versions of treatment"). The notation $Y_j(a)$ presupposes no interference (Cox 1958) and no multiple versions of a (Neyman 1935) (Rubin 1980), plus *consistency* (Robins 1986): $Y_j(a) = Y_j$ when $A_j = a$. These govern randomized trials as well as observational studies.

6.1.3.2 Consistency and well-defined interventions

Consistency: if $X_i = x$ then $Y_i = Y_i(x)$, which requires "$X = x$" to mean the same for everyone. Example: "losing 20 kg" via diet and exercise ($Y(x)$ = survived) versus via starvation or illness ($Y(x)$ = died) is one nominal treatment with different outcomes, so $Y(x)$ is ill-defined. Fix: specify the intervention ("20 kg loss via a supervised diet and exercise program"; *What If*, Fine Point 3.4).

6.1.4 Causal estimands

Motivation. The quantities typically targeted.

Differences.

$$\begin{aligned} \text{ATE:} \quad & \tau = \mathbb{E}[Y(1) - Y(0)], \\ \text{CATE:} \quad & \tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z], \\ \text{ATT:} \quad & \text{ATT} = \mathbb{E}[Y(1) - Y(0) \mid X = 1]. \end{aligned}$$

Ratios. Individual $r_i = Y_i(1)/Y_i(0)$; population $r = \mathbb{E}[Y(1)]/\mathbb{E}[Y(0)]$, not an average of the r_i unless the effect is homogeneous; lift $l = \mathbb{E}[Y(1) - Y(0)]/\mathbb{E}[Y(0)]$. Continuous outcomes dominate the social sciences, binary ones medicine.

6.1.5 Randomization and observational studies

Motivation. Why randomization identifies causal effects, and what replaces it in observational data.

6.1.5.1 Unconfounded assignment

An assignment mechanism is *unconfounded* if it depends only on pre-treatment covariates:

$$p(X = 1 \mid Z, Y(0), Y(1)) = p(X = 1 \mid Z).$$

$p(X = 1 \mid Z)$ is the *propensity score* (Rosenbaum and Rubin 1983). In randomized experiments it is unconfounded and also known.

6.1.5.2 The role of randomization

Randomization balances observed covariates ($X \perp\!\!\!\perp Z$), unobserved ones ($X \perp\!\!\!\perp U$) and potential outcomes, i.e. ignorability

$$X \perp\!\!\!\perp (Y(1), Y(0)).$$

Then, by ignorability and consistency,

$$P(Y(x)) = P(Y(x) \mid X = x) = P(Y \mid X = x), \qquad x \in \{0, 1\},$$

so association is causation:

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1) \mid X = 1] - \mathbb{E}[Y(0) \mid X = 0] = \mathbb{E}[Y \mid X = 1] - \mathbb{E}[Y \mid X = 0].$$

As graphs. Nothing points into X (e.g. $X \to Y \leftarrow Z$). The SWIG splits the treatment node into observed X and intervened x: $X \mid x \to Y(x) \leftarrow Z$; X and $Y(x)$ are d-separated, so $Y(x) \perp\!\!\!\perp X$.

6.1.5.3 Observational studies

The most common assumptions:

1. **No unmeasured confounding:** $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid Z_i$ (unconfoundedness, ignorability, selection on observables, exogeneity, conditional exchangeability): given covariates, X_i is as if randomized.
2. **Positivity (overlap):** $0 < P(X_i = 1 \mid Z_i) < 1$.

Choosing Z is the adjustment-criterion problem. Graphically $Z \to X$, $Z \to Y$, $X \to Y$; in the SWIG $Z \to X$, $Z \to Y(x)$, $x \to Y(x)$, and $Y(x) \perp\!\!\!\perp X \mid Z$ by d-separation.

6.1.5.4 Identification of the ATE

Under positivity and no unmeasured confounding,

$$\begin{aligned}\tau &= \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}_Z\{\mathbb{E}[Y(1) - Y(0) \mid Z]\} = \mathbb{E}_Z\{\mathbb{E}[Y(1) \mid Z] - \mathbb{E}[Y(0) \mid Z]\} \\ &= \mathbb{E}_Z\{\mathbb{E}[Y(1) \mid X = 1, Z] - \mathbb{E}[Y(0) \mid X = 0, Z]\} && \text{(ignorability)} \\ &= \mathbb{E}_Z\{\mathbb{E}[Y \mid X = 1, Z] - \mathbb{E}[Y \mid X = 0, Z]\} && \text{(consistency)},\end{aligned}$$

positivity making the last line defined for every z. With the conditional expectation functions (CEFs)

$$\mu_1(z) = \mathbb{E}[Y(1) \mid Z = z], \qquad \mu_0(z) = \mathbb{E}[Y(0) \mid Z = z],$$

the key step is

$$\underbrace{\mu_1(z)}_{\text{counterfactual}} = \underbrace{\mathbb{E}[Y \mid X = 1, Z = z]}_{\text{observational}}, \qquad \mu_0(z) = \mathbb{E}[Y \mid X = 0, Z = z].$$

6.1.5.5 Regression estimation of the ATE

Fit $\hat{\mu}_1(z)$, $\hat{\mu}_0(z)$, preferably separately in each treatment group; predict both for all units and average the differences:

$$\hat{\tau}_{\text{reg}} = \frac{1}{N} \sum_{i=1}^{N} (\hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i)).$$

6.1.5.6 Summary

Potential outcomes	$Y(1)$, $Y(0)$
Conditional ignorability (back-door)	$Y(1), Y(0) \perp\!\!\!\perp X \mid Z$

Identification: $\mathbb{E}[Y(1)] = \mathbb{E}[\mathbb{E}[Y(1) \mid Z]] = \mathbb{E}[\mathbb{E}[Y(1) \mid X = 1, Z]] = \mathbb{E}[\mathbb{E}[Y \mid X = 1, Z]]$.

6.1.6 Appendix: Positivity and consistency in causal diagrams

Positivity concerns arrows into the treatment node. A deterministic edge $Z \to X$ violates positivity by design; otherwise positivity is implicit in diagrams.

Consistency concerns arrows out of the treatment node. $X \to Y$ needs a well-defined intervention; if $X = x$ can be reached in ways with different effects (diet, exercise, surgery), the effect is ill-defined (*What If*, §6.4).

6.1.7 Appendix: Confounding and exchangeability

- **Marginal exchangeability** $Y(x) \perp\!\!\!\perp X$: no open back-door paths, as in a marginally randomized experiment ($X \to Y$ only).
- **Conditional exchangeability** $Y(x) \perp\!\!\!\perp X \mid Z$: Z blocks all back-door paths, as in a conditionally randomized experiment ($Z \to X \to Y$, $Z \to Y$).
- The back-door criterion implies conditional exchangeability; the converse needs the adjustment criterion (Perković et al. 2018).
- Confounding (structural) = lack of exchangeability (counterfactual) (*What If*, §7.2, Technical Point 7.1).

6.2 Inverse probability weighting and doubly robust estimation

Motivation. Estimate $\mathbb{E}[Y(x)]$ and the ATE under unconfoundedness and positivity, robustly to misspecified nuisance models.

Recall:

(1) unconfoundedness $\{Y(1), Y(0)\} \perp\!\!\!\perp X \mid Z$;

(2) positivity $0 < P(X = 1 \mid Z) < 1$;

(3) hence identification by adjustment, $\mathbb{E}[Y(x)] = \mathbb{E}_Z[\mathbb{E}[Y \mid X = x, Z]]$.

6.2.1 The propensity score

Motivation. A scalar summary of the covariates that suffices for confounding control.

Definition (Propensity score; Rosenbaum and Rubin 1983)

$e(Z) = P(X = 1 \mid Z) = \mathbb{E}[X \mid Z]$, where $Z = (Z_1, \ldots, Z_p)$ collects the p pre-treatment covariates.

Proposition 6.1 (Property 1: balancing)

$X \perp\!\!\!\perp Z \mid e(Z)$. More generally a *balancing score* is any $b(Z)$ with $X \perp\!\!\!\perp Z \mid b(Z)$; $e(Z)$ is one (Z itself trivially is another), and Rosenbaum and Rubin (1983) show that $e(Z)$ is the *coarsest* balancing score.

Proof [Supplement]. It suffices that $P(X = 1 \mid Z, e(Z)) = P(X = 1 \mid e(Z))$. Left: $P(X = 1 \mid Z) = e(Z)$. Right: $\mathbb{E}[\mathbb{E}[X \mid Z] \mid e(Z)] = \mathbb{E}[e(Z) \mid e(Z)] = e(Z)$. ■

Proposition 6.2 (Property 2: unconfoundedness given the PS)

If $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid Z_i$, then $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid e(Z_i)$.

Proof [Supplement].

$$\begin{aligned} P(X = 1 \mid Y(1), Y(0), e(Z)) &= \mathbb{E}\big[P(X = 1 \mid Y(1), Y(0), Z) \mid Y(1), Y(0), e(Z)\big] \\ &= \mathbb{E}\big[e(Z) \mid Y(1), Y(0), e(Z)\big] = e(Z), \end{aligned}$$

free of the potential outcomes. ■

So adjusting for $e(Z)$ removes all covariate-induced bias.

6.2.2 Inverse probability weighting

Motivation. Estimate potential-outcome means by reweighting observed outcomes.

6.2.2.1 The IPW identity

$$\text{ATE} = \mathbb{E}\left[\frac{XY}{e(Z)} - \frac{(1-X)Y}{1-e(Z)}\right].$$

By consistency ($XY = XY(1)$) and unconfoundedness ($X \perp\!\!\!\perp Y(1) \mid Z$),

$$\mathbb{E}\left[\frac{XY}{e(Z)}\right] = \mathbb{E}\left[\frac{\mathbb{E}[XY(1) \mid Z]}{e(Z)}\right] \overset{\text{uncf.}}{=} \mathbb{E}\left[\frac{\mathbb{E}[X \mid Z]\,\mathbb{E}[Y(1) \mid Z]}{e(Z)}\right] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid Z]\big] = \mathbb{E}[Y(1)],$$

and symmetrically for controls: the Horvitz–Thompson (HT) estimator, or IPTW.

6.2.2.2 Estimators

With weights $\hat{w}_1(Z_i) = 1/\hat{e}(Z_i)$ $(X_i = 1)$ and $\hat{w}_0(Z_i) = 1/(1-\hat{e}(Z_i))$ $(X_i = 0)$,

$$\hat{\tau}_{\text{ipw},1} = \frac{1}{N}\left(\sum_{i=1}^{N}\frac{Y_i X_i}{\hat{e}(Z_i)} - \sum_{i=1}^{N}\frac{Y_i(1-X_i)}{1-\hat{e}(Z_i)}\right) = \frac{1}{N}\sum_{i=1}^{N}\{Y_i X_i \hat{w}_1(Z_i) - Y_i(1-X_i)\hat{w}_0(Z_i)\}$$

is unbiased when $\hat{e} = e$.

6.2.2.3 Normalizing the weights

$\mathbb{E}[\mathbb{E}[X \mid Z]/e(Z)] = 1$, but $\frac{1}{N}\sum_i X_i/\hat{e}(Z_i) \neq 1$ in general. Normalizing within each group (divide $\hat{w}_i$ by $\sum_{j:X_j=x}\hat{w}_j$) gives the Hájek estimator:

$$\hat{\tau}_{\text{ipw},2} = \frac{\sum_i Y_i X_i \hat{w}_1(Z_i)}{\sum_i X_i \hat{w}_1(Z_i)} - \frac{\sum_i Y_i(1-X_i)\hat{w}_0(Z_i)}{\sum_i (1-X_i)\hat{w}_0(Z_i)},$$

typically with $V(\hat{\tau}_{\text{ipw},2}) \leq V(\hat{\tau}_{\text{ipw},1})$ (Hirano, Imbens and Ridder 2003).

6.2.2.4 The weighted population

In the IPW-weighted population covariates are balanced:

$$\mathbb{E}\left[\frac{X}{e(Z)}Z\right] = \mathbb{E}\left[\frac{1-X}{1-e(Z)}Z\right] \ (= \mathbb{E}[Z]),$$

since $\mathbb{E}[XZ/e(Z)] = \mathbb{E}[\mathbb{E}[X \mid Z]Z/e(Z)] = \mathbb{E}[Z]$. The less likely a subject was treated, the more of the population it represents.

6.2.2.5 Pros and cons

Advantages: simple, Horvitz–Thompson foundation; no outcome model; target population (ATE, ATT, ...) set by the weights; extends to time-varying treatments and marginal structural models.
Disadvantages: sensitive to PS misspecification; PS near 0 or 1 gives extreme weights, handled ad hoc.

6.2.2.6 Trimming and truncation

- **Symmetric trimming** (Crump et al. 2009): drop units with $\hat{e} \notin [\alpha, 1-\alpha]$, typically $\alpha = 0.1$.
- **Asymmetric trimming** (Stürmer et al. 2010): drop units outside the common PS range, then treated units below the q quantile of the treated PS and controls above the $(1-q)$ quantile of the control PS.
- **Truncation:** set $\hat{e} < \alpha$ to α and $\hat{e} > 1-\alpha$ to $1-\alpha$.

Range restriction thus removes the non-overlapping tails of the untreated (left-heavy) and treated (right-heavy) PS histograms.

6.2.2.7 PS weighting in randomized trials

In RCTs PS weighting (with an estimated PS) adjusts for chance imbalances and improves precision.

6.2.2.8 Weighting versus balancing

IPW balances covariates only *in expectation*. Balancing methods choose w_i directly so that

$$\frac{1}{N_1}\sum_{i=1}^{N} w_i X_i Z_i \approx \frac{1}{N_0}\sum_{i=1}^{N} w_i (1-X_i) Z_i,$$

minimizing a dispersion measure (e.g. entropy) subject to balance: entropy balancing (Hainmueller 2011), CBPS (Imai and Ratkovic 2014), stabilized balancing weights (Zubizarreta 2015). Only the specified moments are balanced: if $Y \sim Z_1 + Z_2 + Z_1 Z_2$ but $Z_1 Z_2$ is not balanced, the result may be badly biased.

6.2.3 Doubly robust estimation

Motivation. Combine outcome regression and IPW so that one correct model suffices.

6.2.3.1 Terminology

- **Correctly specified model:** the truth lies in the assumed class (e.g. $\mathbb{E}[Y \mid X, Z] = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$ fitted with that linear model).
- **Misspecified model:** it does not (e.g. truth non-linear in Z); the fit converges to the best in-class approximation, generally biased.
- **Consistent estimator:** $\hat{\theta}_N \xrightarrow{p} \theta$; correct specification gives consistency (under regularity), misspecification generally not.

6.2.3.2 Two building blocks

By the symmetry $X \leftrightarrow 1-X$, $e \leftrightarrow 1-e$, focus on $\mu_1 = \mathbb{E}[Y(1)]$. Let $m_x(Z) = \mathbb{E}[Y(x) \mid Z]$ ($\mu_x(z)$ in Section 6.1).

- **Outcome regression:** $\hat{\mu}_{1,\text{adj}} = N^{-1}\sum_i \hat{m}_1(Z_i)$, consistent if m_1 is correct.
- **IPW:** $\hat{\mu}_{1,\text{ipw}} = N^{-1}\sum_i Y_i X_i / \hat{e}(Z_i)$, consistent if the PS model is correct, possibly inefficient.

Combined:

$$\hat{\mu}_{1,\text{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right].$$

6.2.3.3 Verification with the true nuisance functions

$$\mathbb{E}\left[\frac{XY}{e(Z)} - \frac{X - e(Z)}{e(Z)} m_1(Z)\right] = \mathbb{E}\left[\frac{XY - Xm_1(Z) + e(Z)m_1(Z)}{e(Z)}\right]$$
$$= \mathbb{E}\left[m_1(Z) + \frac{X}{e(Z)}\{Y - m_1(Z)\}\right] = \mu_1,$$

since $\mathbb{E}[X\{Y(1) - m_1(Z)\}/e(Z)] = \mathbb{E}[\mathbb{E}[Y(1) - m_1(Z) \mid Z]] = 0$ (iterated expectations, $\mathbb{E}[X \mid Z] = e(Z)$).

6.2.3.4 Two equivalent forms

$$\hat{\mu}_{1,\mathrm{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right] \qquad \text{(IPW augmented by OR)}$$
$$= \frac{1}{N}\sum_{i=1}^{N}\left[\hat{m}_1(Z_i) + \frac{X_i\{Y_i - \hat{m}_1(Z_i)\}}{\hat{e}(Z_i)}\right] \qquad \text{(OR augmented by IPW residual correction).}$$

6.2.3.5 What does it estimate?

With arbitrary limits $\hat{e}$, $\hat{m}_1$ and $XY = XY(1)$:

$$\mathbb{E}\left[\frac{XY}{\hat{e}(Z)} - \frac{X - \hat{e}(Z)}{\hat{e}(Z)}\hat{m}_1(Z)\right] = \mathbb{E}\left[Y(1) + \frac{X - \hat{e}(Z)}{\hat{e}(Z)}Y(1) - \frac{X - \hat{e}(Z)}{\hat{e}(Z)}\hat{m}_1(Z)\right]$$
$$= \mathbb{E}[Y(1)] + \underbrace{\mathbb{E}\left[\frac{X - \hat{e}(Z)}{\hat{e}(Z)}\{Y(1) - \hat{m}_1(Z)\}\right]}_{R}.$$

$\hat{\mu}_{1,\mathrm{dr}}$ estimates $\mathbb{E}[Y(1)]$ iff $R = 0$.

Scenario 1: PS wrong, OR right ($\hat{m}_1 = m_1$). Conditioning on Z and using unconfoundedness,

$$R = \mathbb{E}\left[\frac{e(Z) - \hat{e}(Z)}{\hat{e}(Z)}\,\mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z]\right], \qquad \mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z] = m_1(Z) - \hat{m}_1(Z) = 0.$$

Scenario 2: PS right, OR wrong ($\hat{e} = e$).

$$R = \mathbb{E}\left[\frac{e(Z) - \hat{e}(Z)}{\hat{e}(Z)}\,\mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z]\right] = 0 \quad \text{since } e(Z) - \hat{e}(Z) = 0.$$

So $\hat{\mu}_{1,\mathrm{dr}}$ is consistent in both cases (and when both are correct). With $X \to 1 - X$, $e \to 1 - e$, $m_1 \to m_0$,

$$\hat{\mu}_{0,\mathrm{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{(1 - X_i)Y_i}{1 - \hat{e}(Z_i)} + \frac{X_i - \hat{e}(Z_i)}{1 - \hat{e}(Z_i)}\hat{m}_0(Z_i)\right]$$

is consistent for $\mathbb{E}[Y(0)]$.

6.2.3.6 The doubly robust ATE estimator

$$\hat{\tau}_{\mathrm{dr}} = \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right]}_{\hat{\mu}_{1,\mathrm{dr}}}$$
$$- \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left[\frac{(1 - X_i)Y_i}{1 - \hat{e}(Z_i)} + \frac{X_i - \hat{e}(Z_i)}{1 - \hat{e}(Z_i)}\hat{m}_0(Z_i)\right]}_{\hat{\mu}_{0,\mathrm{dr}}}.$$

6.2.3.7 Concluding remarks

- **Double robustness:** $\hat{\tau}_{\text{dr}}$ is consistent if either the PS or the outcome model is correct; a large-sample property.
- Both correct: smaller asymptotic variance than IPW. Outcome model correct: larger asymptotic variance than direct regression.
- Empirically the outcome model matters more: a misspecified $\hat{m}$ gives notable bias even with correct $\hat{e}$, so PS augmentation helps only somewhat.
- Kang and Schafer (2007, *Stat. Sci.*): moderate misspecification of both models (a likely scenario) gives DR large bias and variance.

6.2.3.8 Summary

1. $e(Z) = P(X = 1 \mid Z)$ is a scalar balancing summary of the covariates.
2. IPW reweights by $1/e(Z)$: simple, outcome-model-free; extreme weights.
3. Outcome regression models $\mathbb{E}[Y \mid X, Z]$: efficient when correct; sensitive to misspecification.
4. DR combines both and is consistent if either model is correct.

6.3 Conditional average treatment effects

Motivation. How effects vary with covariates, and how ML meta-learners estimate $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$.

6.3.1 Why heterogeneous treatment effects?

Motivation. A single average can hide opposite subgroup effects.

6.3.1.1 The average can be misleading

20 individuals, half women ($Z = 1$), half men ($Z = 0$), null average effect:

$$P(Y(1) = 1) - P(Y(0) = 1) = 0.5 - 0.5 = 0.$$

Yet the causal risk difference is $0.6 - 0.4 = +0.2$ among women and $0.4 - 0.6 = -0.2$ among men, $\frac{1}{2}(+0.2) + \frac{1}{2}(-0.2) = 0$. *There is generally no such thing as "the causal effect of treatment"; it depends on the population* (Hernán and Robins, *What If*, §4.1).

6.3.1.2 Effect modification

Definition

Z is a *modifier* of the effect of X on Y when the average causal effect of X on Y varies across levels of Z.

- **Additive** effect modification: $\mathbb{E}[Y(1) - Y(0) \mid Z = 1] \neq \mathbb{E}[Y(1) - Y(0) \mid Z = 0]$.
- **Multiplicative** effect modification: $\dfrac{\mathbb{E}[Y(1) \mid Z = 1]}{\mathbb{E}[Y(0) \mid Z = 1]} \neq \dfrac{\mathbb{E}[Y(1) \mid Z = 0]}{\mathbb{E}[Y(0) \mid Z = 0]}$.

Modification can be scale-specific (risks $0.2 \to 0.4$ and $0.4 \to 0.8$: equal ratios, different differences); *qualitative* modification (opposite directions) appears on both scales.

6.3.1.3 Why care?

1. **Policy targeting:** if treatment harms women and helps men, treat only men.
2. **Transportability:** a different modifier distribution in the target population changes the ATE.
3. **Mechanisms:** modification may reveal biological or social mechanisms.

Conditioning on all covariates gives the CATE $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$ (*What If*, §4.3).

6.3.1.4 From stratification to CATE estimation

Stratum-wise effects require discretizing continuous modifiers, suffer the curse of dimensionality, and leave strata with few observations. Meta-learners and causal forests estimate $\tau(z)$ with flexible ML instead (*What If*, §4.2).

6.3.2 Meta-learners for CATE

Motivation. CATE estimators from arbitrary regression algorithms (Künzel et al. 2019, PNAS).

Under unconfoundedness and positivity, $\mu_x(z) := \mathbb{E}[Y(x) \mid Z = z] = \mathbb{E}[Y \mid X = x, Z = z]$, so any regression method applies.

6.3.2.1 T-learner and S-learner

T-learner: fit $\hat{\mu}_1(z)$ on the treated and $\hat{\mu}_0(z)$ on the controls;

$$\hat{\tau}_T(z) = \hat{\mu}_1(z) - \hat{\mu}_0(z), \qquad \widehat{\mathrm{ATE}}_T = \frac{1}{n}\sum_{i=1}^{n} \big(\hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i)\big).$$

S-learner: one model $\hat{\mu}(z, x) \approx \mathbb{E}[Y \mid Z = z, X = x]$ with X as a feature;

$$\hat{\tau}_S(z) = \hat{\mu}(z, 1) - \hat{\mu}(z, 0), \qquad \widehat{\mathrm{ATE}}_S = \frac{1}{n}\sum_{i=1}^{n} \big(\hat{\mu}(Z_i, 1) - \hat{\mu}(Z_i, 0)\big).$$

T is noisy when one arm is small; S may regularize the effect of X towards zero.

6.3.2.2 Regression imputation

Impute only the *unobserved* counterfactual:

$$\hat{\tau}_{RI} = \frac{1}{n}\sum_{i=1}^{n} \Big[\big(X_i Y_i + (1 - X_i)\hat{\mu}_1(Z_i)\big) - \big((1 - X_i)Y_i + X_i \hat{\mu}_0(Z_i)\big)\Big].$$

6.3.2.3 X-learner

1. Estimate $\hat{\mu}_0(z) \approx \mathbb{E}[Y(0) \mid Z = z]$ and $\hat{\mu}_1(z) \approx \mathbb{E}[Y(1) \mid Z = z]$.
2. Impute effects $\tilde{D}_i^1 := Y_i - \hat{\mu}_0(z_i)$ (treated) and $\tilde{D}_i^0 := \hat{\mu}_1(z_i) - Y_i$ (controls); regress to get $\hat{\tau}_1(z) \approx \mathbb{E}[\tilde{D}_i^1 \mid Z = z]$ and $\hat{\tau}_0(z) \approx \mathbb{E}[\tilde{D}_i^0 \mid Z = z]$.

3. Combine with $g(z) \in [0,1]$, e.g. $g(z) = \hat{e}(z)$:

$$\hat{\tau}(z) = g(z)\,\hat{\tau}_0(z) + (1 - g(z))\,\hat{\tau}_1(z).$$

Illustration. One covariate $x \in [-1,1]$, true $\tau = 1$, rare treatment ($g = \hat{e} \approx 0.05$). $\hat{\mu}_0$ (many controls) is flexible with jumps; $\hat{\mu}_1$ (few treated) is nearly linear. $\hat{\tau}_T = \hat{\mu}_1 - \hat{\mu}_0$ inherits the jumps, while $\hat{\tau}_X \approx (1-g)\hat{\tau}_1$ is nearly linear and close to τ, since $\hat{\tau}_1$ uses the accurate $\hat{\mu}_0$ and the effect itself is simple.

6.3.2.4 DR-learner

Apply the DR score of Section 6.2 to $\tau(z)$:

1. Estimate the nuisances $\hat{\mu}_x(z)$ and $\hat{e}(z)$.
2. Construct the pseudo-outcome

$$\tilde{Y}_i = \frac{X_i(Y_i - \hat{\mu}_1(Z_i))}{\hat{e}(Z_i)} - \frac{(1-X_i)(Y_i - \hat{\mu}_0(Z_i))}{1 - \hat{e}(Z_i)} + \hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i).$$

3. Regress $\tilde{Y}_i$ on Z_i flexibly to obtain $\hat{\tau}(z)$.

Why the DR-learner?

- $\mathbb{E}[\tilde{Y}_i \mid Z_i = z] = \tau(z)$ if one nuisance is correct (the DR computation, conditional on Z).
- *Neyman orthogonality:* nuisance errors enter $\hat{\tau}$ only through their product.
- It adapts to the complexity of $\tau(z)$, even with more complex nuisances (Kennedy 2023).
- Use K-fold cross-fitting: nuisances from folds $\neq k$, $\tilde{Y}_i$ on fold k.

6.4 The R-learner

Motivation. A loss whose minimizer is the CATE itself, so nuisance errors matter only at second order.

6.4.1 Motivation and identification

Motivation. The target and its residualised representation.

We observe i.i.d. $O_i = (X_i, Z_i, Y_i)$, $X_i \in \{0,1\}$, and want $\tau(z) = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = z]$. S-, T-, X-learners can pass nuisance bias into $\hat{\tau}$; DR-learners regress an orthogonal pseudo-outcome; the R-learner builds a *loss whose target is $\tau(\cdot)$ itself* (Nie and Wager 2021).

Assumptions: consistency $Y_i = Y_i(X_i)$, unconfoundedness $\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp X_i \mid Z_i$, overlap $0 < e(Z_i) < 1$. **Nuisances:**

$$e(z) = \Pr(X = 1 \mid Z = z), \qquad m(z) = \mathbb{E}[Y \mid Z = z],$$

m being the Z-only mean, not an arm-specific μ_x. With residuals $\tilde{Y}_i = Y_i - m(Z_i)$, $\tilde{X}_i = X_i - e(Z_i)$,

$$\tilde{Y}_i = \tilde{X}_i\,\tau(Z_i) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid Z_i, X_i] = 0,$$

Robinson's partially linear decomposition with a varying coefficient.

Derivation. With $\mu_x(z) = \mathbb{E}[Y(x) \mid Z = z]$, $\tau(z) = \mu_1(z) - \mu_0(z)$, consistency and unconfoundedness give $\mathbb{E}[Y \mid X = x, Z = z] = \mu_0(z) + x\,\tau(z)$ and

$$m(z) = \mathbb{E}[Y \mid Z = z] = e(z)\mu_1(z) + \{1 - e(z)\}\mu_0(z) = \mu_0(z) + e(z)\tau(z),$$

so $\mathbb{E}[\tilde{Y} \mid X, Z = z] = \{X - e(z)\}\tau(z) = \tilde{X}\tau(z)$; the remainder is ε.

6.4.2 The R-loss

Motivation. Turn the representation into a loss and interpret its minimizer.

Oracle R-loss. Knowing m and e, minimize

$$L_R(\tau) = \mathbb{E}\big[\{\tilde{Y}_i - \tilde{X}_i\tau(Z_i)\}^2\big], \qquad \tilde{\tau} = \arg\min_{\tau}\Big\{\frac{1}{n}\sum_{i=1}^{n}\big(\tilde{Y}_i - \tilde{X}_i\tau(Z_i)\big)^2 + \Lambda_n\{\tau(\cdot)\}\Big\},$$

with regularizer Λ_n (Robinson 1988; Nie and Wager 2021, §2).

A residual regression. Given $Z = z$, $Q_z(t) = \mathbb{E}[(\tilde{Y} - \tilde{X}t)^2 \mid Z = z]$ is least squares of $\tilde{Y}$ on $\tilde{X}$ through the origin; $0 = \mathbb{E}[\tilde{X}(\tilde{Y} - \tilde{X}t_z) \mid Z = z]$ gives

$$t_z = \frac{\mathbb{E}[\tilde{X}\tilde{Y} \mid Z = z]}{\mathbb{E}[\tilde{X}^2 \mid Z = z]}.$$

Why this slope is the CATE. $\mathbb{E}[\tilde{X}\tilde{Y} \mid Z = z] = \mathbb{E}[\tilde{X}\,\mathbb{E}[\tilde{Y} \mid X, Z = z] \mid Z = z] = \mathbb{E}[\tilde{X}^2 \mid Z = z]\tau(z)$, so $t_z = \tau(z)$: *the CATE is the slope of outcome residuals on treatment residuals.*

6.4.3 Algorithm

Motivation. Implement the R-learner with cross-fitting.

1. Split the data into K folds.
2. For each fold k, fit on the other folds $\hat{m}^{(-k)}(z) \approx \mathbb{E}[Y \mid Z = z]$, $\hat{e}^{(-k)}(z) \approx \Pr(X = 1 \mid Z = z)$.
3. Out-of-fold residuals: $Y_i^r = Y_i - \hat{m}^{(-k_i)}(Z_i)$, $X_i^r = X_i - \hat{e}^{(-k_i)}(Z_i)$.
4. Estimate $\hat{\tau} = \arg\min_{\tau}\Big\{\frac{1}{n}\sum_{i=1}^{n}\big(Y_i^r - X_i^r\tau(Z_i)\big)^2 + \Lambda_n\{\tau(\cdot)\}\Big\}$.

The R-loss weights residual treatment variation.

$$\sum_i \big(Y_i^r - X_i^r\tau(Z_i)\big)^2 = \sum_i (X_i^r)^2\Big(\frac{Y_i^r}{X_i^r} - \tau(Z_i)\Big)^2.$$

Units with large $|X_i^r|$ (treatment poorly predicted by Z_i) inform $\tau(Z_i)$; those with treatment nearly determined by Z_i barely do.

What can optimize the R-loss? The second stage is empirical risk minimization,

$$\min_{\tau\in\mathcal{F}} \frac{1}{n}\sum_i (Y_i^r - X_i^r\tau(Z_i))^2 + \Lambda_n\{\tau\} :$$

(sparse) linear τ, splines, kernels, or boosting, forests, neural nets (any squared-loss learner with weights $(X_i^r)^2$, targets Y_i^r/X_i^r), tuned by *validation R-loss*, not outcome prediction loss.

6.4.4 Orthogonality and the quasi-oracle property

Motivation. Why nuisance errors matter only at second order.

For a candidate τ use the moment

$$\psi(O;\tau,m,e) = \{Y - m(Z) - (X - e(Z))\tau(Z)\}\{X - e(Z)\}, \qquad \mathbb{E}[\psi(O;\tau^*,m,e) \mid Z = z] = 0$$

Proposition 6.3 (Neyman orthogonality)

Let $\hat{m} = m + \delta_m$, $\hat{e} = e + \delta_e$ and hold $\tau = \tau^*$. Then

$$\mathbb{E}[\psi(O; \tau^*, \hat{m}, \hat{e}) \mid Z = z] = \underbrace{\delta_m(z)\delta_e(z)}_{\text{product}} - \underbrace{\tau^*(z)\delta_e(z)^2}_{\text{square}}.$$

Proof [Supplement]. With $\tilde{Y} = Y - m(Z)$, $\tilde{X} = X - e(Z)$, the moment is $\mathbb{E}[(\tilde{Y} - \tilde{X}\tau^* - \delta_m + \delta_e\tau^*)(\tilde{X} - \delta_e) \mid Z]$. Here $\mathbb{E}[(\tilde{Y} - \tilde{X}\tau^*)\tilde{X} \mid Z] = \mathbb{E}[\varepsilon\tilde{X} \mid Z] = 0$, $-\delta_e\mathbb{E}[\tilde{Y} - \tilde{X}\tau^* \mid Z] = -\delta_e\mathbb{E}[\varepsilon \mid Z] = 0$, and $(-\delta_m + \delta_e\tau^*)\mathbb{E}[\tilde{X} - \delta_e \mid Z] = (-\delta_m + \delta_e\tau^*)(-\delta_e)$ as $\mathbb{E}[\tilde{X} \mid Z] = 0$. ■

No first-order term remains, so flexible nuisance fits are usable (Nie and Wager 2021; Chernozhukov et al. 2018).

The quasi-oracle property. With the R-risk $R(\tau) = \mathbb{E}[\{Y - m(Z) - (X - e(Z))\tau(Z)\}^2]$, $\hat{\tau}$ fitted on $\hat{m}, \hat{e}$ residuals and $\tilde{\tau}$ the oracle, Nie and Wager show

$$R(\hat{\tau}) - R(\tau^*) \approx R(\tilde{\tau}) - R(\tau^*),$$

provided nuisance RMSEs are $o(n^{-1/4})$ (product $o(n^{-1/2})$). The rate is governed by the complexity of τ, not of m and e.

Why cross-fitting matters. If unit i trains $\hat{m}, \hat{e}$, they can memorize its noise, shrinking residuals, and the second stage may read overfitting artefacts as heterogeneity. Out-of-fold residuals avoid this: orthogonality handles nuisance error, cross-fitting prevents reusing the same noise.

6.4.5 Comparison and summary

Motivation. Contrast the R-learner with the DR-learner.

	DR-learner	R-learner
Second stage	regress $\hat{\phi}_i^{DR}$ on Z_i	minimize a residualised weighted loss
Orthogonality	via the EIF / DR score	via Robinson residualisation
Main nuisances	μ_0, μ_1, e	m, e
Model selection	validation MSE for $\hat{\phi}^{DR}$	validation R-loss

DR pseudo-outcome:

$$\hat{\phi}_i^{DR} = \hat{\mu}_1(Z_i) - \hat{\mu}_0(Z_i) + \frac{X_i\{Y_i - \hat{\mu}_1(Z_i)\}}{\hat{e}(Z_i)} - \frac{(1 - X_i)\{Y_i - \hat{\mu}_0(Z_i)\}}{1 - \hat{e}(Z_i)}.$$

Validation losses on a held-out V: DR, $\sum_{i \in V}\{\hat{\phi}_i^{DR} - \hat{\tau}(Z_i)\}^2$; R, $\sum_{i \in V}\{Y_i^r - X_i^r\hat{\tau}(Z_i)\}^2$.

Uses of the R-learner: CATE after removing prognosis $m(z)$ and assignment $e(z)$; model selection by validation R-loss; shrinkage or structure placed directly on τ when the signal is weak.

Summary.

- Start from Robinson's decomposition $Y - m(Z) = \{X - e(Z)\}\tau(Z) + \varepsilon$.
- Cross-fit m and e, then minimize the R-loss.
- The R-loss is orthogonal to nuisance error and fits many ML tools.
- Quasi-oracle: difficulty lies mainly in the complexity of τ.